

ActInf GuestStream #008.1 ~ "Exploring the Predictive Dynamics of Happiness and Well-Being"

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hello everyone and welcome to the active inference lab today it is august 9th 2021 and we're here in guest stream number 8.1 with mark miller and guests so this is going to be an awesome presentation followed by a discussion so if you have any questions during the presentation please put them in the live chat and we'll make sure to ask the panel at the end of the presentation so thanks again to all of our great guests for coming on and please take it away thank you excellent thanks for that daniel okay i'm just going to share my screen oh can you uh turn on screen sharing for me on your side um it's under the advanced so at the bottom where it says share screen there should be a little little arrow and click that there should be an advanced feature and then should say allow got it okay go for it yeah that's it uh-huh great how's that can you just see my slide does that look right excellent great okay well daniel thanks thanks so much man for the invitation to come and give a talk i mean we've been trying to get we've been trying to find an opportunity to speak together for you know a couple of months now well since i moved to japan so i'm excited to be here and i'm excited to be talking about this topic which we're um we're really excited about it's a new paper that actually julian kiverstein who's here right now and myself and eric reitfeld um have a preprint out right now in the the paper title title of predictiveness of happiness and well-being we're also anticipating that um inez hippolyto and laura sinbad smith will be here this evening lars has had a slight emergency so um he's on the road but we're hoping he'll be able to call in um because actually the work that uh we feature in this paper it builds on or it's really closely related with the research that anez and lars are doing so the idea was that we're gonna be able to get um we're gonna be able to really talk to them about their individual portions so hopefully they're on the stream by the time we hit that place and uh if it turns out that they're not that's okay we can just go through and then we can chat at the end um if you don't know me uh my name is mark miller i'm an assistant professor at the university of hokkaido um in their center for human nature artificial intelligence and neuroscience and i am uh an active inference researcher right

so let's see right so i guess probably lots of people who are listening to the stream tonight and definitely the guests who are here right now hey anez great you made it right on um know that um recently there's been a huge amount of research coming out that's linking um using the active inference framework in order to investigate all sorts of psychopathologies you know mostly finding a home in computational psychiatry circles but really we're we're seeing active inference accounts come up you know relating to depression and addiction and ocd and schizophrenia and dissociative disorders and all sorts and while there is this sort of active inference boom right now in relationship to computational psychiatry and in particular psychopathology we're not seeing any work really talking about um how we might apply active inference frameworks to the theoretical and formal approach of discussing things like well-being so you know all those topics that would naturally fit in something like a positive psychology are still primarily being overlooked although you know um in all fairness we are starting to see it come up at the edges because we're starting to see people talk about the benefits of hallucinogens and also the benefits of meditation so that was part of our motivation in writing this paper was to see if there was anything interesting to say in a positive account yeah not just the negative account but the positive account um but the other reason why we decided to write this paper was because actually julian and eric and i have written a series of papers now on psychopathologies in particular using the active inference framework so we have papers out now on um depression and mood disorders and addiction and depersonalization and ocd and we started to see that there were some shared characteristics between these various psychopathologies if you look at them through the sort of computational lens of active inference and seeing those shared characteristics sort of started providing we think clues about you know what might be going right when a human is happy healthy and well and that's what our paper is about and that's what this guest stream will be about as well looking a little bit more at what some of the predictive dynamics might be that are underlying um subjective happiness and overall well-being and of course it's a big topic so we're not trying to exhaust you know and try to exhaust the whole of what well-being would be but rather just taking a sort of first step to see if we can work out some of the some of the points that might be interesting so since i have these lovely people here tonight with me um i've decided to make my presentation relatively short um i hope it won't be any more than sort of 30 minutes all said and um i'm going to pitch it at a relatively high level of abstraction because i think there's like a number of nice moves i'd like to give to the audience here tonight to sort of provoke research and provoke some ideas and um if you're interested more in the

nitty gritty um we have the paper out in pre-print so you're more than welcome to go and actually check out the paper and of course if any questions come up we can talk about that as well great okay so so i think a sort of natural place to start thinking about the relationship between active inference and well-being is to think about optimality um in cognitive neuroscience and psychology today well-being is often spoken of as a form of optimal psychological functioning um but what exactly optimal means is not very well explained usually or at least it's underdeveloped and of course as a free energy principle and active inference framework researcher when you look at that you the first thing i thought was well there's a natural way to sort of be thinking about optimality um in terms of the active inference framework right free energy minimization over the long run approximates optimal active inference right that is sort of bayesian inference is a form of optimal belief updating so that fact in some way already gives us you know it gives the free energy principle in the active inference framework you might think a sort of normativity already you know predictive systems tend to create or strive towards an optimal fit between their phenotypical self-organization and the sort of statistical regularities that they encounter in their environments so um we think that an important part of well-being from this perspective is going to have something to do with being a good predictor of the sort of hidden causes so that's both in the environment and inside the agent of the sort of sensory states that we tend to encounter and of course that crucially relies on our being able to optimally update our generative model in ways that are life-sustaining and flourishing inducing and i don't think that that's an incredible stretch just as an opening gamut um you know we know that well-being is closely related with health in various um various layers of health you know mental health and emotional health and physical health and good predictors they tend to self-organize in ways that allow them to maintain homeostasis across various contexts and over the long term right so um oh and you know and sort of best of all um if we think along these lines which i think could be fruitful lines to think along than we have at our disposal as lots of um the listeners today already know this sort of ever evolving and ever-growing mathematical suite to help us describe what is a good model what's a good model all about so that's our starting point our starting point was to think well maybe well-being could be discussed broadly as a kind of optimal updating yeah and the goodness the goodness of the generative model that the agent uses to form beliefs and control their actions in life-sustaining ways over the long term okay so that's a good start and i think it fits very nicely with what we already know about lots of psychopathologies in particular that psychopathologies are characterized by sub-optimal beliefs and the sorts of

behaviors that come from having sub-optimal beliefs um depression is a good example it's just one out of a hat though really i think if you look at any psychopathology you're going to find some sort of bad belief is at play at some level of the hierarchy i think depression is a nice example you know recent accounts of depression and not just from the act of inference camp um are highlighting the role that rigid beliefs about one's inability to manage the complexities of their life including especially the social complexities of their life is playing a major role in depressive symptoms and we've written a paper on this um pretty recently and um in that paper we using the active inference framework we argued that when an agent is met with enough persistent volatility then they can eventually develop a high level prior about their own high level prior belief about their own um inability to manage volatility and when you get uh that high level prior installed something pretty nasty can happen right um the system a predictive system then starts selectively sampling the world to try to confirm that hypothesis and in fact it down regulates any sort of counter evidence that goes against that deep prior that deep pathological prior so so that even when the situation changes you know if the agent finds themselves in more positively predictable situations uh they tend to hold on to that belief in volatility and that's really the what we think is the pathological moment so another way of saying that is an agent at least a depressed agent yeah has uh become overly confident that it is uncertain it's overly confident about being uncertain we're not the only people who've said that clerking colleagues have written a fantastic paper um definitely check it out if you don't know it and in that they come to the same conclusions they write in this sense we might conjecture that major depression occurs when the brain is certain that it will encounter an uncertain environment okay and um as we all know you know due to the central role that prediction plays in sculpting our experience and directing our behaviors of course as soon as we have a high level bias like this installed um then it has tremendous power to produce exactly the outcomes that would um that would confirm those biases it's one of the problems of being a predictive agent you know the predictive system is really good for lots of things but one of the problems is there's this sort of deep self-fulfilling prophecy that can occur okay the rigidity of these bad beliefs is especially important here and it comes up again when we start thinking about what well-being might be um you know there's mounting evidence that cognitive rigidity you know the sort of stickiness of bad beliefs um but they're a basic feature of psychopathologies and in fact you know um some people are even saying that it might be the root cause of all psychopathologies and um i don't think we disagree um there's a nice aeon article about this if you uh if you're interested in that point and i think i might just uh take a

take a little break there and hand it over to anez um and this hippolyto has a really just a fantastic forthcoming paper coming out that everyone should be excited to be reading which offers um a really beautiful computationally rich account of how insulated internal states yeah when these bad beliefs sort of get stuck and then they resist updating how that creates um the sorts of pathological belief systems the pathological behaviors that we see in psychopathology so um if it's okay and as i'm just gonna stop share and let you maybe talk a little bit about insulated bad beliefs and then i'll pick it up from here and keep going on to well-being you know is that okay yes for sure um hi everyone can you hear me and see me yeah okay great so first of all thanks so much for mark for inviting me to be here and it's always such a pleasure to be here with daniel uh for these wonderful sessions and uh hi julian uh very nice to see you as well so i'll just now share my screen if i think that's possible yeah let me see okay so for some reason i'm not able to share my screen i think it should maybe try again okay no it doesn't go through for some reason uh strange it works for me fine i have no idea we could go for a uh an audio thought or you can try to share it later let me just leave and come back if that's okay maybe that's good you know the trick um this is uh this is this is the new world of technology isn't it you have to do everything on the fly you have to do everything on the fly and be willing to adapt i i have one interim question yeah please yeah i hit it were you working on active inference or interested in those approaches and then came to be studying psychopathology or just well-being more broadly or was it the other direction like which way did you come at this intersection from yeah you're right and um it started the other way around so um in my masters in which julian was my advisor at the university of edinburgh in 2010 um the last chapter of my senior thesis was a proposal so i was working on empathy looking at um um you know philosophy and cognitive science of empathy looking at dynamic systems theories of emotion the last chapter was on predictive processing and active inference and then i got to do my phd with andy clark specifically on active inference and then it grew out of doing that into psychopathology yeah we got it very good we got it excellent we got it some technical difficulties but uh we are now good okay all right you can see it fine yeah yep nice work thanks and let's continue perfect okay so um all right so um yes this work is on uh uh what we call psychotic macro blankets and this is a something that i've been working on uh for the last few months with two different groups applying this to uh different um uh two different things so one specifically schizophrenia specifically and the other one is um disembodiment so what i thought i would do here today would be to do like sort of like a combination between um the two so that we get like a bigger picture of what uh we have in mind with them with uh

with applying this particular framework so um yeah so um let me just uh give some like framing about uh where this comes from and uh why and to motivate it a little bit so that was the parts from this concept that mark also just mentioned the self-organization which is super relevant to these things that we are talking about when we refer to well-being and mental health and all of that um and as a few words about that it's an emergence of order from disorder and specifically without an intervention from the external intervention that is external to the system um and then what is important here for us is to think of self-organization as a pattern or an arrangement that gives rise to a specific way in which the system is self-producing and self-distinguishing and here and not to the inactivist literature so but the idea here that the fundamental concept is the self-organizing pattern that that comes up this is why why are we talking about that because we want to look at how the systems or the organisms as a whole are interacting with the environment in a way that has been called in embodied literature as a tournament with the environment right so the idea there is that these organisms or the systems are a tune with their environment and uh in a way that they will show a certain specific pattern um or part of the arrangement that will be distinct from for example systems that are not attuned to the environment and this is very well known in the whole bulk of literature in this embodiment and phenomenology so uh the idea that we want to pursue here is actually relating to a little bit too complex systems theory in which we want to say that this differentiation between embodiment and disembodiment are different patterns it doesn't mean that the system that the system that is not adjusted to the environment should be considered as disordered but instead that it has a different kind of order right so he would be sort of like rejecting a little bit the computationalist accounts of psychopathology as simply a disorder a disorder or an illness and opening up a little bit of more like a broad scope or spectrum of um to talk about mental health um so then if we set up things like that then uh what needs to be explained is how uh this self-organizing behavior of situated organisms come to be ordered in a certain way rather than another and then we'll be uh now talking about uh mental health so um then we think that active inference is um a very interesting and useful tool precisely to think about uh this uh self-organization and the patterns that come from the self-organization um we think about um active inference as a model of uh leaving things adjust to the environment and we all know what active inference is so i'm just gonna highlight what is relevant here for us in terms of pattern dynamics we think that it is a nice and cool model that when associated with tools such as macro blankets or renormalization group it can be applied to many different scales of self-organizing activity and that's where we find all of these

patterns of our interest of our scientific interest so it is useful to explain um these gradients spectrums of kinds of patterns of order in self-organizing systems so just highlighting these ideas um so we take it as a model um in this case so it means that we are not committing to the to the realist view of that let's say that the phenomenon studied under active inference ontologically possess or leverages the conceptual tool but that the tool is really important and useful for us to understand uh certain phenomena in this case particularly what you want to understand is this embodiment right so we all know active inference but again let me just highlight uh just some important points to understand than the psychotic markov blankets later on here the idea is that self-organizing self-producing self-distinguishing system will correspond to the internal states which affect and which effects and affect and are affected by the environment and the external states via a set of other states such as the active and the sensory states active and sensory states are then constructed as influencing one another giving rise to a certain pattern of self-organization and this is what's going to be relevant for later on so in a way what we want to say is that well an adjusted system that self-organizes is a system that um minimizes the free energy which we all know can also be understood as uncertainty or entropy and so on right but an adjusted system would be something like that a system that is just well adjusted to its environment and for technicalities obviously there are these references on the bottom now moving on to this embodiment and then later on applying we want to apply this active inference strategy to further highlight or understand what is going on with this embodiment um just a few words about the disembodiment literature so the idea is that comes from phenomenology and a little bit from existentialist philosophy and the idea is that the primary experience of the body is the body experienced as a subject and this is really crucial the thing is that this experience of the body as a subject can be lost and in which case the body is inspected as an object more like a side of scientific interest and here is what uh the literature says that that's when this embodiment comes up is when the body all of a sudden ceases to be experienced as a subject and is now inspected as an object so what happens is uh in exploring and interacting with the local environment we usually typically we do not attend to our body we just like take it for granted got our body that allows us to access move around interact explore the world through our body and it is not until we cannot do these things or something is wrong that we will direct our awareness towards the body and then the body becomes salient as an object of inquiry and it's right here at this point where we find this embodiment it's when we cannot um explore or interact with the world as one would like would like so in a way we can um distinguish embodiment and disembodiment by saying uh for example or just

highlighting these features of course the literature is vast but i think that these are the ones that work for us here um embodiment is an attunement with the environment that is allowed or enabled by having an experience of the body as a subject um and that gets us attuned um to the environment through a bodily self-awareness or something that there is a bodily self-awareness that is that is uh felt then this embodiment on the other on the other hand is a disatnment with the environment because there is a loss of the body as a subject body becomes an there's a disatnment and a diminished body awareness so this comes from the and it is interesting for us here why because then there's been these links that have been made between psychopathology and psychedelics and disembodiment as in this embodiment being something that is reported in different psychopathology such as schizophrenia the personalization even major depression or other psychopathologies that involve rumination and another very important thing is that this embodiment also occurs or is what is reported in the encyclopedias intake and this has been um even more relevant uh studying this embodiment precisely in the psychedelic renaissance that we are witnessing today and also another point that i want to make uh just here sounds like in brackets is just that disembodiment does not necessarily need to be a bad thing right we usually put things in good or bad right or wrong but it doesn't need to be a bad thing because if we do get some therapeutical effects um positive effects from psychedelics this precisely by having this disembodied experience so it's just a different way of interacting with the world and that's why we want to pursue less of a disorder way and much more of a there's a spectrum of different patterns of interaction with the world okay so just that note um then um i want to look at this embodiment um and how we can use active inference to understand this embodiment in two ways one is the psycho the psychology of this embodiment so how that disappointment is felt psychologically and the other one is the neurobiology of this embodiment so as we have said uh this embodiment is the loss of the body as a subject which is replaced by uh some form of intellectual effort so imagine that you now don't have these broadly access to the world that is not possible because that's exactly what has been lost by psychopathology or psychedelics intake and many other things right so then try to cope and compensate by literally engaging in making inferences about the state of affairs so here we could use a little bit of the literature and sophisticated active inference which is this conscious reasoning and planning and policy selection that you engage now with having lost these access to the world through the body now our hypothesis is that the issue is that in this disembodied because we all engage in sophisticated active inference that's all fine um the difficulty is that in the case of this embodiment um experience

this inference is insulated is isolated because what has been lost is the body access to the world so then what happens is um so uh we would have something that looks like this so we all know this um this scheme here right and what we want to say is that what is lost now is the influences between sensory and active states and let me guide you through it so um let's just uh focus on the figure for now so we have internal states as being disembodied reasoning so a subject that has lost this access or bodily access to the world because the body is not felt as a subject so we've got this disembodied reasoning as internal states and then we have external states as the environment where that that the subject wants to explore and interact with and then we have a sensory state the environmental co-constraints so these aspects that should influence uh the subject right and then as active straight states we have the effectivity as um the ways in which the subject can affect and being effect and be affected by the environment but what has been lost here is that there is an unbalance between sensory and active states so for one active states are not sensitive to sensoristics it is this effectivity or this possibility of being affected by the environment that is not possible why the access through the sub the body as a subject has been lost in this psychopathology or psychedelics as that we label as disembodiment experience so um as a coping mechanism what is going to happen is there will be an increase of activity of the activity of active states by increasing the issuing of inferences about the world so i do not understand what is going on but i'm going to continue to think about it and make and and develop theories about what is going on the problem is that we can we all engage in these developing theories and inferences about what we think is the likelihood of the state of affairs the situation here in this embodiment is that these inferences are disembodied so they tend to be false beliefs and that's because of the insulation of internal states what does this mean this means that there is not um so the access that we had with the world through our lived body is what is lost so then the subject in a situation where um the only hope is to increase in increase active states to make more inferences about the world to cope with that loss and then tendency is to have false beliefs that's when we start entering the realm of psychopathology and psychosis so in these disembodied cases uh this is actually a very important point because one could say okay so then the problem is the is on the level of reasoning so on the level of for example of the internal states as we have set up here no the problem is not on the level of the internal states reasoning is fine the inference mechanisms are working perfectly fine the subject can engage in in reasoning it can make inferences the problem is inside the markov blanket the problem is in the unbalance that there is between active and sensory states because those are the ones that are supposed to directly influence one

another such that indirectly there's a connection between internal and external states but it is this um direct influence between sensory and active states that is problematic they are not communicating well they are not influencing uh themselves or each other in a balanced way um in a sense and balanced in the sense that active states become super active and sensory states um cannot influence active states so then what we get we get to this idea of a psychotic markov blanket which is if we have this sort of arrangement where there's a problem between the balance of the strength of states then we have a problem of the generation of false beliefs the problem being that these beliefs that are issued by active states they cannot be updated by the world because it is precisely this access to the world that has been lost lost by virtue of having lost the body as a subject so then um because of this uh this achievement that we can actually explain as what is appearing inside the marco blanket the reasoning subject makes inference about inferences about world but um these reasoning occurs from within the lack of the experience of the body as a subject so it occurs in an insulated manner and this is consistent actually um with a long-standing literature and phenomenology of psychosis um where uh the subjects and sisters a bunch of children on psychosis linked to solipsistic behavior um solid existing uh drawing and painting and also the philosophical idea which is the philosophical idea that only one's mind is sure to exist and then some some reports that i find quite interesting is uh uh some patients who report i was no longer sure that i was still the same person i became one with other creatures or objects i lost a sense of my own boundaries when i had an experience i often did not know it was mine or the experience of someone else so it is quite interesting what can be lost when we lose these body as a subject as it is developed in phenomenology this also happens a little bit in the personalization disorder um for example showing this disembodiment is achievement when making people feel estranged and detached from their self body and the world okay so this is all i had on the disembodied on the psychology of this embodiment and now i'd like to turn to the neurobiology and also in the end apply the same framework now to the neurobiology of um of the underlying neurobiology of this environment so we think that applying the same reasoning is is consistent so this this achievement that you were talking about felt at agent level should also be expected at the neurological state so that's our hypothesis right so now we think that there are two places there are two good candidates for perception and interception would be good candidates um super perception as a system underlying movement and location of the body in the environment as well as interception as the system responsible for homeostasis these are good candidates we think um for the neurobiology biology of this general disorder that is felt in this embodiment

so proprioceptive and interceptive systems um just like any situated organism can be conceived as self-organizing systems so we apply exactly the same principle as we would apply to the agent as a self-organizing system so the question then that we had is what is the factor that is at the basis of this of a neurobiological pattern underlying proprioception and interception that will show this small adjustment to the local environment so if we expect the same sort of mechanism or organization dynamics as to the whole agent and that we were just talking about then how are we going to find this out as well in the neurobiology so we think that just like in psychological experience of this embodiment where the agent becomes insulated from the world and in this case we say by virtue of losing the body as a subject the answer will reside in biological insulation um now i'm going to focus now on interception because i don't have time to also cover proprioception um and i think that this does give us here some very very cool insights so um i'm going to uh focus now on the work uh brought by baritone siemens on interceptive predictions in the brain and they have a vertical model which they call embodied predictive interception coding model and well the goal there is to integrate an anatomical model of corticocortical connections with beijing active inference principles and they propose a very cool thing which is that a granular visceral motor cortexes contribute to interception by issuing interception predictions so more specifically what they want to say is that the granular limbic regions that regulate visceral motor of the body's um internal milia that is the visceral motor cortexes so this is their focus and where do you think they will find the interceptive uh system and where they apply the model now um so you can see the agranular cortex here uh on the left and uh one very interesting thing that i'm just gonna draw your attention to and then if you feel uh that you'd like to read this super interesting paper then um please go for it but i just want to draw your attention to the fact that on the left side you have the a granular cortex and there's one thing that is missing from the a-granular cortex as opposed to the these granular and the granular cortex so what is missing is the layer 4 and what happens with that is that there won't be you cannot see there any prediction error neurons so that's pretty cool because that will uh give us an idea that there is no prediction error neurons in the a granular cortex um so just i'm just going to leave that um out there and um just moving uh forward to just like a understanding like the very cheese of the embodied predictive interceptive coding model um they claim that there is an interceptive system in the brain in which a granular cortices sends visceral motor predictions to the body and transmits interceptive predictions about the visceral sensory consequences of those predictions so they also tell us that adrenal are visceral motor cortexes estimate the balance between the

autonomic metabolic and immunological resources that are available to the body and the predicted requirements of the body based on past experience right so this is going to be very important for us a chronic imbalance on those things that i was just mentioning can be which can be caused by constantly predicting the need for more metabolic energy to meet the demands for stressors can produce the well-known depression related disruption and it can also eventually down regulate provoke a down regulation of hypothalamus or hpa axis negative feedback loops resulting in chronic hyper cortisolemia and in turn can also promote pro-inflammatory state so this is like some of like the problems of having an imbalance on the level of the autonomic metabolic and immunological resources which is what the interoceptive system should be able to do is to give us that nice balance right so then applying exactly the same um the same framework the same model that we applied to the whole body this embodiment or the generalized feel of this embodiment that is vastly described in the phenomenology literature applying this to the neurobiology of the interoceptive system then what we get is something that can be seen as like quite similar an unbalance between sensory and active state so let me just guide through the figures so here in this figure internal states are interests are or correspond to the interoceptive system and the external stage the environment whichever um the multi-scaled system that receptive system is part of and then we have active stages that a granular quartus that send visceral motor predictions and then we have sensory stage stages the requirements of the body given the history that the body knows to be required for homeostasis so then in this sense what we would have is that active states are not sensitive to sensory states so even though active states can become uh as a coping mechanism much more active and send much more predictions about what is going on what's the state of affairs um because i'm not receiving any information from the requirements of the body so the sensory states so what will happen then because of this insulation is that a false belief is going to be generated so this will correspond to a constantly prediction of the need for more metabolic energy to meet the demands of the stressors so in this sense we have here another case of what we call a psychotic markov blanket why is it psychotic because it delivers if there is this lack or this imbalance between sensory states or the influences between sensory states and active states then what we should expect is a degeneration of a false belief and we can also expect that at a neurobiological level which is also supported by empirical evidence so yeah this is what i wanted to say uh just thinking um thanking these uh collaborators here that have been thinking about and guiding me through as well these ideas thank you ness really interesting stuff so i guess we'll return to mark and then anybody is yep anyone's free to ask a

question in the live chat and i'm writing things down and then after this second piece of the presentation we'll have a discussion so thanks again go for it mark yeah that was great thank you so much wow it's so interesting and i haven't heard some of your stuff about disembodiment yet and you know also this is a research program that i'm extremely interested in so uh i'll be picking your brain at the end of the q a and also after after this event um what might what wasn't made obvious in anez's talk but i think is implied is that the stickiness the stickiness of these belief systems and what's helping them become insulated is a story about how precision is being estimated and how precision is being deployed that's surely one of the mechanisms that is creating an imbalance between active and sensory states so you know how precision is estimated is really the agent's ability to assess the uncertainty of its own priors and when that's working right then those estimations should result in appropriate model updating you know you shouldn't get stuck in an insulated in an insulated situation but rather you should know wow that's not working out the right way and so appropriate updating should occur um and we know that like maladaptive precision estimation is disastrous right and we're seeing it everywhere in the psychopathology literature today you know schizophrenic delusion is being um characterized as too little precision on sensory data which is putting this heavy emphasis on just top-down predictions so that's why you're seeing the world in a way that isn't being anchored by what's really happening and of course asd is a little bit the flip right too much precision on prediction errors creates a lot of a lot of volatility and a little bit difficulty um establishing good high level predictions and as a quick side note you know the the kind of insulated insulated um outcomes that create these maladaptive uh self-organizing styles they can occur in lots of ways right so for instance yeah so addictive substances is one good way right addictive substances impact directly precision estimation machinery right in the way that the addictive substance engages the dopaminergic system so addictive substances are setting precision outright in a way so no wonder they can create these sort of sub-optimal styles but societal expectations do the same sort of thing you know sub optimally pinning precision on expectations about success for example you know um so you can have cultural pinning of precision that you should achieve a certain level of success in your life and you know we have deep running relationships between perfectionism and depression for example and again for exactly the same reason that anez said overly high precision on some sort of expectation would lead to a constant pervasive background error as you're not hitting that high level of expectation and then just like um feldman barrett suggests in simmons you have an hpa flip because the system just doesn't have unlimited resources um and then the one that i'm most

interested in today actually is social media which is turning out to be a like fantastic way of bending our generative model into all sorts of shapes that cause sub-optimal outcomes right and that makes sense it's just giving us wave after wave of um you know misinformation through filtered images and staged images and of course social media platforms some social media platforms function just like addictive substances or like casinos in a way so again they're hijacking the precision system in just the sorts of ways that cause these problems okay i just want to quickly summarize before i go on to why what i think the natural mechanisms that keep us out of these problems are so well-being as optimal model generation importantly depends on correct and flexible yeah not these stuck insulated outcomes but correct and flexible self-estimations of uncertainty so that's precision assignments um so misassigning precision to prediction errors which can happen in lots of ways especially in our in our modern world um will inevitably lead to sub-optimal styles of self-organization sub-optimal generative models and with that the production of pathological beliefs and pathological behaviors and so that's how it can go wrong or at least that's the starting point it's one of the ways that it can go wrong so that's how it goes wrong then what is it that ordinarily keeps us out of these sorts of bad bootstraps these sorts of insulated uh disharmonies between the the system and the environment and one answer that i like that we like as a team it was given by alex chance and colleagues er where they said that optimal free energy minimizing agents will benefit from striking a good balance between pragmatic and epistemic engagements okay so that is they don't only act in a pragmatic way that is in ways that they're highly confident will reduce free energy over time because they can be wrong that's part of the take-home message from some of the pathological self-organization styles right they say that you should also a good free energy minimizing system is going to balance that against being driven to epistemically explore and forage for more information about how the world works but also about how their own model works and implied there is that when they find important counter evidence while they're out exploring that they're willing to update their model relative to that counter evidence so then the question then if we buy that and i i think i do buy that's one of the ways that we stay out of these bad boot straps and these sub-optimal styles of interacting well how do we find that right balance i think that's a good question if we're gonna ask well how do we live well what is well-being all about how do we find that good balance and the central way that we've been arguing over a number of papers is that we literally feel our way to that balance and um that part of course is predicated on everything working right because if you have an affective disorder then it's going to be exactly that mechanism that's going to be mis-tuned and

of course we get all the problems um that anna's just pointed out so um in a series of recent papers we've been exploring the suggestion that um good precision assignment is going to rely in part on the rate at which error is being reduced so that's commonly referred to as aerodynamics and um we're increasingly not alone in this conjecture um we have people like jeff lee and corcelli who gave the first paper on this and of course vanda cruz and then it was beautifully formalized by casper hesp and colleagues um so if you want to look really at the nitty gritty of this that's definitely the paper to check out and if you want to see a great paper on the computational bits of this you should see casper's talk it's there on the slide it's fantastic so um the idea um uh the idea has developed that these aerodynamics yeah the changes in the rate at which error is reduced are registered by the organism as affective states and um if we think of an agent's performance in reducing error in terms of a slope yeah that plots the various speeds that prediction error is being accommodated relative to our expectations then we can think that positively and negatively balanced affective states are a reflection of better than or worse than slopes of error reduction so um that is just to say we feel good when we're doing better than expected at reducing error relative to expectations yeah better than expected and we feel bad when we do worse than expected at reducing um prediction error and these embodied aerodynamics are important drivers of precision estimation it turns out and i think that makes sense of course right because unexpected increases or decreases in volatility relative to your model is important information for how you should set precision on your beliefs about action policies right so unexpected decreases in the rate at which error is being reduced tells you that your action policy isn't working the way that you thought it was going to work which means you should obviously reduce your confidence in that action policy and the other way as well right if you're unexpectedly doing better at reducing prediction error relative to some policy then you should up your confidence in that policy so the point being here is that precision then isn't adjusted only based on the amount of error or the amount of error reduction but precision is also being tuned relative to the rates at which error is being minimized over time and um that feature of the active inference framework gives us a rather sexy account of momentary subjective happiness which i think is so interesting right um momentary subjective happiness is the result of unexpectedly reducing prediction error and that feels good because we are suddenly doing better than we thought we would at getting a good predictive grip on our environment which under normal circumstances would usually mean that we are staying healthy and we're staying well and um juffley and corchelli's graph makes that clear you know as free energy is going up um you feel fear and then you feel unhappy when it's

super high and then you feel hope when it starts getting better and you feel happy when uh error is being managed in an extremely good way but actually it's not only showing up in active inference these ideas are being presented in um in other camps as well rob rutledge is of course the paradigmatic case um he and his team have done a number of uh brain imaging studies that have already started to show the strong relationship between subjective feelings of happiness and doing better than expected in some particular nation you can see on the board behind rob there that's the calculation for happiness which i think is a bit of a cheeky way to go but i think it's cool um and uh if you know the maths then of course you're gonna you're gonna know what's going on there too now we get to the fun stuff this is the last thing this is sort of we're coming into the last stretch here's the fun stuff so we think that the same machinery that underlies momentary subjective happiness is also going to play a leading role in helping us tend towards optimality and to avoid these sort of sub-optimal styles of self-organization so we think it's going to play a special role in an active inference account of what it means to be well um to see why aerodynamics play that role we just have to ask ourselves well where are the best slopes to be found and the answer is the best slopes are found at the edge of our own predictive capabilities right between what's known and highly reliable and what's unknown and potentially more optimal and when it's all working right aerodynamics naturally move us out to that edge right where errors are neither so complex that we can't learn anything from them nor too easily predictable that there's no more epistemic value and really the best slope is going to be right there at the edge of your own capabilities and you know that's not a provocative that's not a provocative idea we see that everywhere in the curiosity literature um we know that babies are attracted to medium amounts of complexity not too well known not too volatile we see it in mice put them in a maze with varying rooms of uh grades of complexity and the mouse will tend to hang out in the room that's just above its comfortability and um and you know we're already building um adaptive and effective robots using these ideas so this isn't this isn't a far out idea so important for our current discussion is that in order to make the most of those good slopes that are right at the edge of our own capability a predictive system must be sometimes willing to disrupt its own fixed point attractors yeah so it must be willing to go beyond its habitual policies to go beyond its extremely well-known ways of self-organizing beyond its own homeostatic set points even right and that's one of the roles that aerodynamics should play when everything is working right when it's going well and you're on a good slope of error reduction then you should be motivated to continue along that path but when the niche thesis to give you productive

prediction errors then the negative valence that comes up that signals to the system that it's time to destroy some of these fixed point attractors in favor of more wandering policies more explorative styles and i mean if you want to see that in technical terms um just check out uh Fristen breaks beer and deco that's the paper to go to so agents that use aerodynamics in this way to set precision on action policies they're going to avoid getting stuck in any one fixed attractor basin for too long right they'll instead tend to exist in more meta-stable states more often they're going to be poised they're going to be optimally poised in some way between exploiting already existing off action policies and performing action uh information seeking epistemic actions that reduce uncertainty and that metastable poise actually grants all sorts of fitness advantages over strictly ordered or more chaotic systems at least in our kind of environment and that's something that's very well known in the dynamic systems literature so again this isn't a very provocative claim here um and it's precisely because those sorts of systems have an optimal balance between you know what you might think of as efficiency and degeneracy another way of saying that is um they're set up to respond efficiently um to particular contexts those contexts that are highly predictable but they're also remaining open to exploring and growing in a wide variety of other possibilities okay so notice then that um for a free energy minimizing system in our kind of flourishing is not about avoiding errors i know that's starting to be more popular but i think it's a really good point to like hit right on the head it doesn't flourish by avoiding all it flourishes by selectively sampling the right errors and i think that's so cool like if that wasn't immediately obvious if you're still not like you're still not really catching that point then i think it's it's a good point to catch soon um what matters for flourishing for a free environment is that it gets the right kinds of errors because the right kind of errors into it and of course we know that you know like outside of the model it only sounds weird i think inside the model if you're in that camp outside of the model we all we already know that you know overly sterilized environments make children sick overly simplified cities make people depressed like we know too little volatility um causes problems for complex self-organizing systems like us again i don't think it's such a provocative claim although it can sound provocative from inside the active inference framework maybe so finding the right sorts of volatility and the suite of cognitive emotional mechanisms that keep us in touch with that good volatility uh we think that that's going to play center stage in any account of well-being if we're going to look at it through the active inference lens and we've done a little bit of work on this already this is our aeon article where we do this in a sort of popular way um if you want to check it out the title of the article is the value of uncertainty good okay uh last little bit and

then we're done oh um in and now we're going to come into the most speculative part of the idea and so then we'll stop and then we can we can talk about it and see where it goes so in the literature on well-being um researchers tend to congregate in a couple of big camps um and two of the most popular camps are thinking about well-being in terms of feeling good that's the hedonic camp and people who think that well-being is about being good and that's the eudaimonia camp and actually those go all the way back to aristotle featured here as the cool guy in the glasses so we speculate in our paper that actually we might be able to map those two subjective momentary subjective happiness and overall flourishing being a good sort of predictor on two forms of aerodynamics a local form of aerodynamics and a global form of aerodynamics and we think that there's an uh good reason to start thinking about how we might distinguish between local and global aerodynamics um because for instance think about video game design okay so a well-crafted video game could be highly tailored in such a way that it provides you with just the right evolving changes to keep you continually making learning progress relative to the games you're going to keep finding those good slopes and we all know you know good game design does do that it keeps creating these good slopes of air reduction so do good casinos right um but um just like with substance addiction right you can get so immersed in those local good slopes that um you end up surfing them at the expense of the other things that matter in your life so you start neglecting your friendships and your schoolwork and your work um and your overall fitness okay so while you're succeeding locally at the game uh you're not flourishing overall because in fact there's going to be rising volatility across all of these other domains that you're temporarily not paying attention to right so sensitivity to global aerodynamics would essentially amount to being driven in ways that reflect how well we're managing errors across the various domains in our life and as long as an agent was using those more global estimations of how well or poorly they're doing over their various concerns uh to adjust their precision then you're going to have an agent who tends to stay in touch and who tends to move towards optimality relative to those multiple cares and concerns so you're not only improving locally but we think that there's good reason to think that we're also sensitive to how we're improving overall and that's going to play a special role in well-being in particular and um that makes if that's right if we're right here then that makes a really natural fit between this computational work that we've been interested in and more recent so-called network accounts of well-being where well-being is thought of as the result of tending to a complex ecosystem of positive states and positive achievements so then global aerodynamics would be crucial here we think to staying in touch with those various concerns

in the kinds of way that would allow you as a whole to tend towards a good overall optimal shape um now like i said this is the most speculative part of the talk nobody has talked yet about thinking about aerodynamics happening in a nested way but i just want to say that we do have good reason to be thinking along that line it's unfortunate that lars isn't here with us tonight because it piggybacks on some of his deep parametric modeling so i was hoping to lay it off on him to really get into the computational nitty-gritty of why we're not just pulling this out of a hat for you um but we so we build up this idea that he's already been developing um and the idea is basically um he uses a deep parametric model and you can check out his paper on metacognition um if there's a way that we can um link this somewhere i can yeah we can okay i'll send it to you afterwards um but he ends up showing that you know there's higher level policies that are governing the allocation of precision on lower level tasks and the kind of cool thing about this deep parametric model is that it becomes possible to appreciate precisions over those higher level policies over precision allocation to particular tasks so you get this sort of nested you get this nested precision estimation where at the lower level you have precision being adjusted relative to a particular task at a higher level you have precision adjustments that are also effective we suspect uh being set over the policies about how you deploy your precisions over all the things you care about so that that better than or worse than the global better than or worse then is going to be all about are you doing better than or worse than a deploying precision across your various cares and concerns not relative to some local concern so then we think that what breaks down an addiction in particular is you start paying attention to aerodynamics locally in the drug seeking and taking behaviors and you actually have a separation of context and we see that in the brain right you see the striatum pull apart from the dorsolateral prefrontal cortex so the more contextualizing parts of the brain are no longer communicating in the same sorts of ways they were before because you have ultra high precision on these local aerodynamics so they're good again is a good example where we're going to get an insulated sort of psychotic markov blanket emerge because of an overestimation of precision because the the drugs of addiction are making that happen so lars and myself are currently working on clarifying that aspect of the model so look forward to work soon on that um and we're looking at how we might apply that model to thinking about um meditative development thinking about attending to attention precision on precision modeling precision um and uh thinking about how those sorts of mechanisms might help us have better access to and be better guided by these higher level precision tuning processes um yeah so look forward to that and uh maybe that'll be another thing we can come on and uh talk

about here on the lab and then hopefully lars will be uh not an emergency situation that time and he can come on great um so that's probably enough for me up maybe just as a final note before we turn it over to q a i might just say um the future directions for this are plenty if you buy some of the mechanisms that we've been thinking about right it would be cool to start now thinking about what sorts of endeavors encourage or disrupt that sort of meta-stable poise right meditation hallucinogens provocative art installations um you know are they are they helping or hindering um can we foster these metastable states you know and and how you so deep brain stimulation looks like it's doing something like this you're getting stuck in a tractor basin an ocd deep brain stimulation and julian can talk about that if it's an interesting point reopens up the field of affordances by creating more dynamic neural um ecosystems what does this mean for how we structure our environment if this is right if we flourish by hanging out at the edge of our own capabilities then what does that mean for how we structure our environment how should we build our cities i mean abby tabber at um at bath is doing incredible work on this thinking about how we might structure our cities in ways that are good for us and actually they turn out to be sort of paradoxically not making the city easier for us but making it more challenging getting more diversity more art installations more guerrilla gardening uh things because actually the more complexity we have actually the better we are as individuals and of course the biggest thing i'm working on right now here at uh hokkaido university is what does that mean for how we design our technology and how we consume our technology how about you know what role is misinformation as a digital crowbar uh prying us apart from the statistical realities of our environment what role is that playing if these flexible fluid dynamics are part of what it is to be well that's all that's all future stuff and uh if you're listening to this tonight and you think it's a cool talk and there's something of value here then uh that's the way i suggest starting to think next and i'd love to hear anybody you know here on q a but also um my door is always open if you uh if you sort of dig the talk and you dig the research then definitely reach out because we're we're always up for collaboration that's probably good for me you're welcome awesome so fun and so many cool ideas in there um anyone in the chat is welcome to provide a question and first i'll just pass to julian to maybe give any remarks or thoughts and then we'll see if there's any questions in the chat or some questions i've written down no i think mark did a great job so let's just jump straight into the q a thanks mark awesome um okay so you mentioned it kind of tantalizingly at the end about how we would differentially be using the technology that just made me wonder like what would inactive inference social media look like or how would somebody of course not see any examples of it

out in the wild how would we even go about imagining that kind of a system anybody else want to try to answer first because i've got something to say if anything comes up should i just should i just go yeah yeah i mean it's a good question it's something i'm really interested in um maybe one thing to say is uh you might think that uh there's a big market for making tech that uh reduces our option set um you know um that oh well this tech is great for us because it simplifies our affordance landscape and that's what people want they want a sort of simplified affordance landscape and the danger is is that actually we're going to be attracted to that because it's going to temporarily reduce free energy faster than we thought it would right google does a great job of quickly minimizing our uncertainty about various topics but over time uh you can have a sort of nasty outcome and i think we're starting to see that more and more where you collapse because it's so rewarding to come to one place like you know your smartphone ends up being one place to come again and again and again and over time your affordance landscape is collapsing it's increasingly collapsing to one and only one end it's a little bit like walmart you know it you know you want to buy there because you can get great deals but the problem is is that when the whole community buys there and all the mom and shop complexities go out of business uh then you're stuck just with walmart and then they put up their prices it's a little bit you know it's sort of like that you think that the collapsing uh tech is going to be good for us but actually over time having less complexity is harmful for us and think about social media platforms as well like you sterilize social interaction and it's really rewarding at first because social interaction is hard it's a complex affair you know it's very uncertain how to manage it to be with another human so temporarily online interactions can be really really rewarding because they're so much simplified but if you hang out there for too long and you lose some of your ability to engage with complex human affairs um then we're starting to see some of the ramifications of that sort of technology i mean we're literally dying of loneliness even though we're more superficially connected than we've ever been and i think that would be an interesting project is to start thinking about what is working when we have complex ecosystems of social interactions that we're losing by just using social media platforms for our social interaction yeah julian yeah so i was thinking about uh intellectual humility as this is something we've been talking about before and how important it is to be open to prediction errors if you're going to get the most out of social media so you can think of filter bubbles as a kind of psychotic markov blanket in a way and that there is insulation from any evidence that doesn't match with your your predictions or match with your expectations so in order to flourish to to live well with social media perhaps what's necessary is is this kind of

openness to uh other contradictory points of view uh what the ancients would have thought of as humility and we need to foster that to be well with technologies so inis what do you think about filter bubbles as an example of a psychotic markov blanket do you think that's a stretch or does it fit with your the way that you describe this this insulation from sensory evidence that happens that can induce a kind of solipsism yeah absolutely i think it's uh exactly on the right track um actually uh the way that algorithms are constructed to deliver precisely these bubbles in social media it's so interesting because it's precisely what's going to happen a psychological markov blanket because it's just going to be insulated from that amount of information making inferences about that that precisely precise information without having any other contact or exchange outside of the bubble so that that's very interesting as well and there are many many other forms of psychotic markov blankets that you can find in the societies and communities and um that kind of thing but i didn't want to address exciting things about the concept that it's scale free so uh now you see it at collective levels of self-organization not just but also the level of the cell you show so that nmda receptor you can also think of in terms of a yeah psychology yeah yes i didn't have time to present that here uh because uh yeah but um but that's that's also formulated under the disconnection hypothesis by carl so it's also really cool because it's at a cellular level so it's quite interesting if anyone is interested so that would be also what we are um applying or looking at in the in this paper is another example of a psychotic markup blanket where there's a an imbalance between um again what i was saying the sensory in the active states formulated there um as the receptor so and specifically for schizophrenia but then you can scale all of these up and find all of these psychotic markov blankets um in many different levels and i think that in the case of social media the bubbles are an excellent example but i think that we also have that as mark was mentioning and i always i always find it's very interesting uh when we discuss this topic about um about how um specifically um the city living as an environmental daily living space influence mental health and and specifically why and when some groups of individuals thrive in an urban setting so this is something that some people here in amsterdam for example are working on and it's really interesting um and i think that it becomes even more interesting if we come for example from studies in neurodiversity for example in autism and that kind of thing and if we think about the design of these spaces um daily spaces daily living spaces that um are not designed for example for diverse ways of interacting with worlds such as an autistic interacting with the world so i think that um it is very relevant to think these mental problems and how we can inform interventions in the city planning and also smart

environments um that would allow a more diverse way of interacting with the environment as opposed to reducing the complexity of the environment which is exactly what mark was talking about and i just wanted to add that um to that yeah we're actually working with a pretty cool group through the nested minds network um developing models of anti-fragility and what we're calling trophophilia which is the love of uncertainty so you're not just growing from volatility but you're actually seeking volatility in order to grow and we're thinking about it at the level of communities so um we're still just getting started but um it's a really fascinating really fascinating field yeah so one comment on that and then a question from the chat um julian you pointed out that active inference as a scale free and sort of uh base system agnostic framework really helps draw out these similarities across systems and then mark in the in the presentation you talked about neural ecosystems about social social media ecosystems and then thinking about some of these pathologies or perceived pathologies in an ecosystem case i mean imagine if someone said i like this tree this should be the only tree or i'm seeing a little tension here with a predator and the prey we should separate those two species those would lead to a catastrophic collapse because it wouldn't be the frat it wouldn't be the anti-fragile ecosystem that was the one that had made it all the way to the present moment and was existing far from equilibrium so it's sort of like when we can look across systems it gives us a new way to look at the systems that we can modify the nisha of like our cities the internet and then starts hopefully shining some light on where we can look here's the question from the chat dean wrote remark a simple provisioning rule when in doubt zoom in comma zoom out so that's dean's providing that is sort of like a little simple uh motto as an attending strategy could this be a happy person setting up a local or global non-isolating markov blanket like how do we go from these mathematical framings and some of these aerodynamics into nice slogans like dean does no i love that i mean right yeah exactly yeah look look close and look far i mean that's that sounds like that sounds like super good advice um a little bit you know this a little bit follows on from the same point julian was making earlier about virtue about some of the like what would the virtues be relative to this kind of good updating system and um i've been meaning to do a i've been meaning to write something on this for a long time now and you know i'm getting more and more motivated as we go along but have you ever heard of super forecasters have you heard of this it's a book out called super forecasters so there's people around the world who go to competitions and they predict things from noisy data sets and you can train in it and you get in a team you can do individual events and people can get better so you can train to be a super forecaster so you take noisy data you get good at

predicting it and i like this idea because i think it's a nice metaphor you know i think we might be able to glean like if we're all predictors and we're trying to predict well then maybe there's something to learn from people who try to predict well as a hobby you know and try to see like what are they doing right um and there's some really interesting things that they do uh that i think are um useful given the kind of framework that we're proposing here a couple of them i think just everybody should know about one a super forecaster across the board the best the best super forecasters in the world they share a couple of common traits they are curious so they're always interested not just in their own field but they're interested in lots of field they tend to be polymaths so they're interested in lots of different sorts of things that's one thing the second thing and this is a little bit i feel like is related to this zoom in zoom out point and also exactly what julian said is that they have they have they hold so this active inference language but they hold high-level beliefs that all of their beliefs are only relatively right they know that when they're looking at something not to get too enamored by it because it's up for updating at any time so they're always checking it against the other things they know they're the kind of person that when you meet them in public and you tell them something they don't know they don't lock down and say no no i can't be like that no no you know they're the kind of person who says wow wow i didn't know that tell me more about that so i can update i can update my overall model relative to how the world is anyway i think i think these like this this approach to zooming in and zooming out sounds just right and i think it's so cool to start thinking about what some what are some of the practical virtues that would help us not get stuck and i think uh yes follow up on that so yeah i think what you just described to the super forecasters is something that goes back to the ancients again the peronean skeptics who put forward the idea that we shouldn't do we shouldn't endorse a proposition but nor should we withhold our endorsement we should rather try to find this balance and what you're describing with the super forecasters with that kind of hybrid project i think is exactly that same kind of balance well that's the paper peyronian skepticism through the lens of active inference that's realm it made me think about skepticism and like almost like a bayesian stoicism like okay the info has come in through the channel it came in from i would like to learn more i will marginally update like that sort of approach and then also another dialectic that you brought up that it'd be interesting to hear a little more about is that hedonic short-term happiness versus that um the eudemonia the longer term happiness just like that's super fascinating to see how you mapped two pillars of philosophy really going so far back integrate them in in a new way like where do we go from there how do we think about these kinds of cool synthesis between philosophy and

computational dynamics i love that julie do you want to say anything on that if not i could maybe start and you could follow on what do you think yeah so uh the way we were thinking about it was though to take i don't know if mark did this example in the talks i had to pop out but the computer game example did you go through that mark yes exactly you know you might you might find a a computer game that gives you just the right amount of challenges to keep you constantly making progress in playing the game but still in the rest of the person's life going to school uh having your friendships grow your family life all of these things could be suffering they're just finding local pleasure which we can assume the computer game might continually give the person by presenting them with this constant challenge in the end wouldn't be conducive to flourishing for that person because all of the other aspects of their life that are also valuable and important to them would end up being neglected so thinking about that from a dynamical or a computational perspective it seems what's important is a kind of psychological flexibility and that you're able to be poised between exploiting and exploring so the person who's playing the computer game is really just stuck in a kind of exploitative mode where that's continually yielding something rewarding for them because they're continually presented with challenges that they can solve but at the same time they're not exploring they're not curious and so there's not that kind of psychological flexibility which we think is required for eudaimonia or really living a good life and flourishing as a person and so we think what you get out of active inference then is is a neuro computational frameworks a biological story but also a computational story about uh what it might be to to as a person live a good life what might be required is this kind of openness and flexibility yeah yeah it's great two other quick things on the tale of that one is you know there's a very old debate about higher and lower hedonia yeah the low hedonia is the sort of animal life that causes all the problems higher hedonia but there is higher hedonia that leads you to whatever the good life is that's kind of interesting from this perspective as well because you can think about it i think maybe as lower hedonia is aerodynamics in niches that lead you away from the sort of metastable attunement right so drugs of addiction is a good example of lower hedonia because it's good prediction error slopes but they're only local at the expense of your more integrated holistic optimality of governing state right but what would a higher hedonia be well it would be a local good slope that leads you to a greater integration so at the tail end of our paper and i didn't mention it here because it's a bit of a further topic but we suggested maybe thinking about zero sum and non-zero sum activities and their role in producing or restricting these sorts of metastable attunements and i think that's a good example where you can get you can

get hot about some small thing that reduces your dynamicism so you get this global rising in air even though you're locally succeeding that's a nice i think it's a nice way to think about lower hedonia higher hedonias local good slopes that encourage and expand and and generate a richer and more complex uh ecosystem of good slopes in your life so for instance like well like think about the things that are endlessly rewarding you know love relationships you know like you're never going to get to the end of being a good husband or wife you're never going to get to the end of being a good older brother or being a good dad good mom you're never going to get to the end of those that's that's an endless thing or patience patience compassion love you know sympathy empathy i mean you're never going to get to the end of those things service service to the people into your community service is such a good example here if you feel good like if you get good slopes in serving your community you're just going to keep creating more connections that are going to create more opportunities for new kinds of slopes to be caught and encouraged so i think it's it's perfect almost you know it's a local slope that leads to lots more other stuff and just one other quick thing and i won't i won't i won't harp on about it but uh and i think also that uh a system that's set up to be uh to have this more global aerodynamic regime working correctly i think they'll also tend to have more subjective and it would be good to like go through this and build a detailed computational model because this part is just a little bit speculative but it seems to me that if you have a really rich diverse ecosystem of good slopes then if you reach the end of some particular slope like let's say you lose the video game if the video game is the only thing you have no wonder we have a problem regulating our emotions if you know you see this on youtube all the time you know mom or dad turns off the system and there's an explosion and they say so this is the only thing i have right so you're very very rigid you're very fragile with only one outcome but if you have lots of stuff going on in your life then if you lose a good slope here it's just a matter of task switching over to the next good slope and so i tend to think that people who have a diversity of good slopes in their life of course they're also going to have more persistent more pervasive good feelings but you know that's a bit speculative so love the look this local global nested slopes descent i think it really helps us understand active inference but also understands or helps us understand how to apply active inference because it doesn't have to be an active inference model about emotion but that's the one we're talking about but it could be some other kind of active inference model so kind of just two other um concordances i guess uh cezanne mahali's flow and flow psychology it's like you mentioned selecting the right kind of errors and so it's like there's there's being in the flow state of playing chess or of doing a skill

and then there's like getting the rock out of your shoe it's like you're you're still in this sort of uncertainty regime but one of those is just delaying you on the journey and the other one kind of is the journey so that and then one other it's dead on it's dead on actually john verveigh at the university of toronto and i are planning a paper using this exact model to think about flow as cascading cascading insight so every time you reduce you hit that good slope and then you learn that's what the good slope is you're learning and when you get to the end of that learning what it does is it opens up a new good slope that you learn down which opens up a new good slope which you learn down and that happens in a sort of cascading a cascading pattern yeah i love that so that that perfectly comes to the second example the cascading insight like so many people whether they're in academia or just curious about something when they get a quote answer they say well you know just brought up so many more questions like now i know how much this planet is in diameter but now i'm curious about something else and so it's like that that short term minimization is like disciplinary research like we had 25 efficiency on this chemistry extraction we got the 27 it's a local who could who can say that's not a local improvement but then that global is the transdisciplinary understanding and as the research ecosystem if we just dove down local disciplinary rabbit holes we would end up with just a wreck but at the same time we do need to do some local gradient descent otherwise we're not extracting that compound so what can we build our transdisciplinary theory on and so it's just like very cool how we can um have a positive um i don't know can it be neutral and positive at the same time yeah yeah why not yeah it's positive in the sense that it's not negative uh i but i i love that and i think that's a i think it's a role that philosophers are increasingly um playing you know what you might call synthetic philosophers um philosophers who are trained in various scientific disciplines the neighboring scientific disciplines around a certain topic like dan dennett you know who are able to zoom out and create good overall frameworks that sort of uh that bring together various disciplines as well as the special sciences that zoom in that zoom in and uh discover locally they come in on this because i think one of the interesting thing is the three of us here are all philosophers so we will come at active inference through philosophy but through trying to do that synthetic kind of philosophy where we're integrating philosophy with neuroscience with biology so that's a common theme and another thing that i think is interesting among the three of us is that we're all thinking about active influence as something that is environment involving so one of the crucial interesting things i think in innis's work on the psychotic mark off blankets is what happens when you are you're you have a a free energy minimizing system that's insulated or isolated from what's coming in from the

environment and how that can induce pathologies and a lot of the things we've been talking about today are about kind of designing an ecosystem being in that kind of environment would help you to flourish as an individual so what would uh how would you need social media to look if it was instead of creating the kind of social pathologies that we're seeing today with the misinformation problems and all of the problems that are arising from polarization you had a social media environment in which people flourish and so we think that these ideas about active inference if you have a philosophical perspective on them you can see how they they could help you to know to think about ethical questions maybe political questions as well it's about synthetic perspective that philosophy can bring helps you to see how to take this framework and show how it can be applied in lots of different domains i think that was something that you touched on daniel in one of your observations questions awesome man i think it's a valuable contribution that philosophers can make from from using these ideas totally we see advances in philosophy alongside and deeply interwoven with governance and technology so here's just one last question i wanted to ask on that theme unless anyone has any other questions and this is um to and as his section so you talked about thinking about using active inference as a model which was invoking this realism instrumentalism distinction that we had talked about in some other streams and i just wondered how will realism versus instrumentalism or potentially other philosophical ideas play out clinically like what will it look like to deploy something instrumentally and say that for example this is being supported by a mathematical framework but that's not what is per se happening inside of you it's just how we're instrumentally interacting with it i don't know just how will that kind of a philosophical nuance enter into the regular protocols of medicine or governance um yeah um so i think that um instrumentalism is very much aligned with dynamical systems theory take on cognition or on brain activity right so it doesn't need to be something is highly mystical or that is like a framework that we use but then what can we do with it really like in in medicine or in more experimental settings right it's just that it's not um committed to a more computationalist account of cognition or of the brain it's much more aligned with for example in this framework with the dynamical systems theory as in the case of being a tool that is useful to explain and describe um an activity of scientific interests through some constructs that we know are constructs that are we build for our own scientific practice and understanding right so that's the idea it is useful in the sense that you can you can now put things together and make this activity that would otherwise be a blur intelligible in a scientific sense by using these constructs which you can think about it for example from dynamical systems theory mark was

mentioning these attractors for example which by the way um i'd like to say something about that um um in the case of um for example in psychopathology there is this kind of like fixed points or attractors that kind of like we need to change etc we are not saying that there is actually a fixed point or an attractor in the psychopathology but we are explaining this in this sense for our own or to make these patterns intelligible from a scientific point of view right so it's in this sense that you can use this instrumentalist account to say that you gain explanatory attraction but you're not saying that actually there is a fixed point in the behavior itself does this help yeah i think so i mean if someone looks at an economic model and it uses a regression it's not saying the economy is a regression but then there's something about the the personal nature of the brain and our emotions where it feels almost like a model of emotion or emotional dynamics it's something that's been brought up many times i mean a lot of people like qualitative perspectives on these topics because it it averts this whole discussion on whether the mathematical models are real or instrumental and it almost feels like now with that philosophical clarity to be able to frame the two different modes we have there we can distinguish very clearly instrumentalism from a sort of background of implicit realism and then that would maybe help speak to a broader set of people who have different beliefs and different structures for what they think is the underlying system can i comment on this briefly yeah please yes i i think i take a different perspective reminisce on the realism instrumentalism issue so i i start from the cybernetic idea of the good regulator theorem that ashby put forward and i think that gives us a way of thinking about how values are actually built into living systems and so the idea of being a model of the environment it's not just having a model but actually being a model and how you can be a good model so what does it take to be a good model while there has to be as much complexity in the model as there is in the environment that you're interacting with so i think what we get out of there out of ashby's good regulated theorem is an idea about what it is to be well adapted to the environment and i don't take that just to be like a model of uh of life or of systems that are well adapted but that we can actually get some principles that tell us how would a system need to be organized in order to be well adapted um and those principles yeah i mean they are a model i agree but they actually tell us something organization that a complex adaptive system would need to have if it was to remain well adapted over time so if it was to be a good model and then we can get some kind of value or normativity out of that because notice that there's a notion of goodness implied here that we're thinking about what would it take for a model to be a good model of its environment for it to remain well adapted when there's a kind of value or normativity implied there as well so i

think all of that is not just there about you know finding explanatory models that do a good job for us as scientists but it's also telling us something about what it is to be alive a good cognizer so that we could begin to get some kind of values out of biology and i take that to be a kind of realist project you know i think that we can we can get some motion or value ourselves free energy minimization in fact i think the notion of free energy is itself a kind of value for a system that tells a a system what it is to and to be well adapted it's well adapted if it if it is first of all minimizing free energy over time but we saw from marx talking from our paper that on its own isn't sufficient you need in addition this kind of maintaining less stable poise uh so now all of that is just to say well i think we can get some something more than just a model out of free energy free energy principle we can get some principles which help us to make sense of how there could be values in a in a world of facts and so it helps us with that question it may help us to understand you know what it is to live a good life so and how could you have natural systems that flourish so we could get some kind of um naturalistic account of uh of ethical values now so that would be my take on it and i think that's kind of different from from instrumentalism a different kind of perspective on instrumentalism i think you can also get out for free energy principles there's a debate here isn't there and that's one of the things that philosophy can bring in there's different perspectives you can have on what the free energy principle means and then we can fight about that amongst ourselves as philosophers great closing point we respect and value the uh the content but then also there's that hyper philosophical prior that the space for the debate and the space for different perspectives is itself important so with that thanks to all of you this was an awesome session you're always as well as anyone else welcome to come back share any other exciting research or questions great times and i hope everyone watching live and replay appreciate this great session so thanks again everyone and see you next time thank you for having us thanks so much daniel thanks everyone thanks so much